

Azka Ikramullah

AI RESEARCHER · GENERATIVE AI ENGINEER

Seoul Metropolitan Area, South Korea

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Summary

AI researcher and machine learning engineer specializing in **large language models, generative AI, and AI safety**. My work focuses on **LLM robustness, reasoning limitations, and adversarial vulnerabilities**, developing empirical evaluation frameworks to analyze failure modes of generative models. I design and deploy **LLM-based systems** including RAG assistants, AI automation tools, and scalable machine learning pipelines. My goal is to advance **trustworthy and secure AI systems** that bridge research innovation with real-world deployment.

Education

Gachon University

Seongnam, South Korea

M.S. IN COMPUTER ENGINEERING

Mar. 2025 – Present

- Graduate Research Assistant at ISML Lab contributing to a Korean government-funded project on LLM security and AI systems.
- Presented research at AAAI and participated in major AI conferences including AAAI and ICML.
- Graduate coursework: Reinforcement Learning, NLP, Computer Vision, and advanced AI systems.

Air University

Islamabad, Pakistan

B.S. IN INFORMATION TECHNOLOGY

May 2019 – May 2023

- Final Year Project: PTSD prediction from textual data using machine learning (97.56% accuracy) with 10Pearls mentorship.
- Selected for Dawlance Women in Tech Graduate Trainee Program focused on applied AI and machine learning.
- Research Internship at National Cybercrime Forensics Lab (Air University / Pakistan Air Force) on ML for audio forensics.
- Coursework: Artificial Intelligence, Machine Learning, Data Science, and Software Engineering.

Publications

Azka Ikramullah, Abdul Majeed, Kyunghyun Lee, Seong Oun Hwang

When Reasoning Collapses: A Depth-Aware Probe of LLM Robustness (Student Abstract)

Singapore

AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE

2026

- Proposes a depth-aware framework to diagnose reasoning failures in large language models.
- Analyzes robustness degradation across multi-step reasoning tasks. [\[Paper\]](#) [\[Code\]](#)

Azka Ikramullah, Kyunghyun Lee, Abdul Majeed, Seong Oun Hwang

SAC-Triad: Reinforcement Learning Framework for Evaluating LLM Jailbreak Robustness

Singapore

AAAI TRUSTAGENT WORKSHOP

2026

- Introduces a tri-agent reinforcement learning framework to automatically evaluate jailbreak attacks on LLMs.
- Enables scalable adversarial testing and safety evaluation of generative AI systems. [\[Paper\]](#) [\[Code\]](#)

Azka Ikramullah, Seong Oun Hwang

More Thinking, Less Reading? Separating Reasoning Utility from Evidence Selection in Factual QA

Sydney, Australia

NEURIPS 2026 – WOMEN IN MACHINE LEARNING (WiML) WORKSHOP

2026

- Accepted for poster presentation at the Women in Machine Learning (WiML) Workshop at NeurIPS 2026.
- Investigates the distinct roles of reasoning and evidence selection in factual question answering, examining when additional reasoning improves performance and when better evidence selection is more important. [\[Paper\]](#)

Technical Skills

Languages

Python, Swift

ML Frameworks

PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers

LLM Systems

RAG, Prompt Engineering, LLM Evaluation, AI Safety, Agentic AI

Tools

Pandas, NumPy, Git

Work Experience

ISML Lab, Gachon University (ISML)

Seongnam, South Korea

GRADUATE RESEARCH ASSISTANT

Mar 2025 – Present

- Developed an evaluation framework for analyzing jailbreak robustness and safety vulnerabilities in large language models.
- Designed a reasoning analysis framework to investigate failure modes and robustness degradation in multi-step LLM reasoning.
- Conducting empirical studies on adversarial robustness and security risks of generative AI systems.
- Exploring multimodal AI systems, RL-based LLM alignment, and agent-based AI architectures.

AIO App Inc. (AIO) AI PROCESS AUTOMATION ENGINEER • Built RAG-based AI assistants for real-time querying of business data through natural language interfaces. • Developed generative AI applications including a GPT-4-powered multi-specialty medical chatbot. • Designed automated testing pipelines for LLM applications to improve response quality and model reliability. • Optimized AI pipelines and Django APIs for scalable deployment, reducing cloud infrastructure costs by 60%.	Islamabad, Pakistan
	May 2024 – Mar 2025
FitFlex (FitFlex) IOS DEVELOPER • Developed Swift-based iOS applications following Apple UX guidelines for production deployment. • Built a machine learning-based message classification system improving prediction accuracy by 70%. • Contributed to an AdTech analytics system for user call behavior prediction and marketing optimization. • Managed cross-functional development tasks to deliver mobile products for client deployments.	Islamabad, Pakistan
	Sep 2023 – Mar 2024

Projects

Multi-Specialty Medical Chatbot

GPT-4, FASTAPI, GENERATIVE AI	2025
• Developed an LLM chatbot delivering responses across cardiology, psychiatry, and dentistry domains. • Implemented specialty-aware prompting and FastAPI backend for scalable AI interactions.	

SOP Generator

GPT-4, PROMPT ENGINEERING	2025
• Generates customized Statements of Purpose from academic profiles and goals. [GitHub] • Implemented structured prompt pipelines using GPT-4 to produce coherent academic narratives.	

AI Code Assistant API

LLM APIS, DEVELOPER TOOLS	2025
• Built API for code generation, debugging, explanation, and image-to-code transformation using GPT models. • Designed endpoints enabling integration with developer tools and IDE workflows.	

Skin Treatment Visualizer

STABLE DIFFUSION, COMPUTER VISION	2024
• Built a web system simulating aesthetic procedures (Botox, fillers) using diffusion image generation. • Allows users to preview treatment outcomes from uploaded facial images.	

PTSD Prediction Tool

MACHINE LEARNING, NLP	2023
• Developed an NLP model detecting PTSD from clinical text records. • Achieved 97.56% classification accuracy on hospital dataset.	

Academic Engagement & Extracurricular Activity

Women in Machine Learning Symposium @ ICML 2026

POSTER PRESENTER	Seoul, South Korea
	Jul 2026
• Presented <i>When Reasoning Collapses</i> on depth-aware LLM reasoning. [Poster] • Analyzed reasoning degradation as inference depth increased.	

AAAI Conference on Artificial Intelligence, ICML

CONFERENCE ATTENDEE	International
	2025–2026
• Attended leading AI conferences and engaged with research on LLM safety, reinforcement learning, and trustworthy AI. • Participated in workshops and technical sessions on generative AI evaluation and AI security.	

MENA Machine Learning Winter School

PARTICIPANT	QCRI, Doha, Qatar
	Feb 2025
• Selected participant in workshops on Responsible AI, large language models, and federated learning. • Engaged with international researchers through lectures and collaborative technical discussions.	

MENAML Conference

APPLICATION REVIEWER	KAUST, Saudi Arabia
	Jan 2026
• Reviewed applications for the MENAML conference selection process. • Evaluated candidate profiles from global machine learning research communities.	

Brain Korea (BK21) Government-Funded AI Research Project

RESEARCH CONTRIBUTOR	Gachon University, South Korea
	2025–Present
• Contributing to a Korean government-funded BK21 research project on trustworthy AI systems and large language model security. • Collaborating with lab researchers on evaluation frameworks for LLM robustness and adversarial safety analysis.	